

Rohith Pallamreddy

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science, GPA: 4.0

Atlanta, GA

May 2027

EXPERIENCE

Incoming at General Motors Cruise (AV Platform OS and CPU team)

Warren, MI

Autonomous Vehicle Software Engineer Intern

Summer 2026

- Will develop and **optimize embedded software platform** components including drivers, operating system, BSP, and software stack for high-performance SoCs powering ADAS and **autonomous driving applications with C++ and RTOS principles**.
- Will work on real-time embedded platform solutions to improve performance, reliability, and **scalability of the core software stack** for production autonomous vehicle systems.

Resilient and Equitable Mobility Analytics and Planning @ Georgia Tech

Atlanta, GA

Machine Learning Researcher (Mapping and Mobility)

Jan 2025 – Present

- Generated spatiotemporal processing pipelines to efficiently compress and extract **70+ GB** of data into **<1 GB**
- Architected **algorithmic trajectory reconstruction** pipelines using Kalman filters and **machine learning techniques**, achieving **96.3% accuracy** with noisy GPS data
- Generated predictive Motif pipelines with **graph-based models** using machine learning techniques to identify and predict mobility-related key human **decision-making** features with **84% accuracy**

Perception and Simulation Engineer (Microclimate)

Aug 2025 - Present

- Processed **500+ GB of LiDAR datasets** to recreate 3-dimensional terrain, building, and canopy information for downstream processing
- Utilized LiDAR derived features to implement SOLWEIG (a physics-based model) to **simulate and predict** perceived temperature with 1m spatial resolution and an **RMSE lower than 1.5° celsius**
- Engineered and debugged alternative SOLWEIG pipeline to simulate and predict perceived temperature with a **93% faster runtime, no spatial compression, and 99.3% accuracy**

PROJECTS

LADEE Lunar Dust Risk Mapping & Path Optimization

Nov 2025 – Dec 2025

- Developed a **high-fidelity Monte Carlo simulation** using **PyBullet** to model 10^5+ dust particles, implementing a danger coefficient that reduced sensor interference risks by 62%.
- Optimized graph-based trajectory planning via **Dijkstra's Algorithm**, improving path safety metrics by 42% across 200+ simulated lunar terrain edge cases.

Motorcycle HIL Simulator

Aug 2025 – Dec 2025

- Developed a Hardware-in-the-Loop (**HIL**) system integrating Arduino-based IMU sensors with a **custom UE4 physics model** to achieve 98% fidelity in real-time lean angle replication.
- Implemented a **Kalman Filter** to fuse noisy accelerometer and gyroscope data, reducing signal variance by 62% and ensuring stable $< 10ms$ latency for high-speed simulation dynamics.

CounterPunch (Computer Vision Defense Analyzer)

May 2025 – Sept 2025

- Architected a real-time perception system using **MediaPipe** to map 3D "strike zones," identifying defensive guard vulnerabilities by calculating spatial coverage gaps in the boxer's stance with **74% accuracy**.
- Engineered a predictive heuristic engine with **90%+ accuracy on beginners** to track head-center coordinates, calculating velocity and acceleration vectors to generate a "Predictability Score" for future movement patterns.

TECHNICAL SKILLS

Languages: C++20 (STL, Boost, Templates, RAII, GDB), Python (C-Extensions, Cython), C, Rust, SQL, Bash
Systems & Sim: High-Fidelity Physics (PyBullet, UE4), HIL/SIL Testing, ROS2 (Pub/Sub, ActionLib), Gazebo
Perception & Math: Computer Vision (OpenCV, MediaPipe), State Estimation (Kalman/EKF, SLAM), Eigen, PCL
ML & Optimization: PyTorch (TensorRT, CUDA Kernels), Scikit-Learn, Distributed Training, Latency Optimization
Tooling & Infra: Linux Kernel/Systems, Bazel (Hermetic Builds), Docker, CI/CD, Git, GCP, Valgrind, Perf